ISE 5405: Optimization I

Lecture 1: Course Introduction

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August 25, 2026

Grado Department of Industrial & Systems Engineering
Virginia Tech

First lectures presented online by Dr. Robert Hildebrand.

Today

- What this course is about: optimization, the science of better decisions.
- A first linear programming (LP) model and a peek at its geometry.
- Real applications and their linear, discrete, and nonlinear flavors.
- The axioms that make a problem linear.
- Why clever algorithms are necessary: brute force fails spectacularly.
- Logistics: materials, homework, exams, and policies.

Three strands, every week

Modeling, algorithms, and analysis (including proofs).

Course Logistics

- **Course:** ISE 5405 — Optimization I, Fall 2026.
- **Meetings:** Tuesdays and Thursdays, 9:30–10:45 a.m.
- **Location:** Durham Hall 261.
- **Canvas:** `https://canvas.vt.edu/courses/236227` for announcements, assignments, and course materials.

Primary textbook

D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, 1997.

Instructional Team

Instructor

Dr. Jiaxiang "Jason" Li — `jasonljx@vt.edu`
Durham Hall 213 — Office hours: TBD

Dr. Robert Hildebrand presents the August 25 and August 27 meetings online.

Teaching Assistant

Contact information and office hours are forthcoming. Once announced, direct solver and Python setup questions to the TA first.

Please ask early and often

Ask in class, during office hours, or through the Canvas discussion channel.

Optimization: Taking Optimal Decisions

In complex systems, "common sense" strategies can be misleading. Optimization gives a principled way to choose the best action subject to constraints and trade-offs.

Every optimization problem has three ingredients

- **Decisions**: what we control.
- **Constraints**: what limits us.
- **Objective**: what "best" means.

Our job this semester is to make these notions precise — and computable.

Where Optimization Runs the World

The same decision–constraint–objective pattern appears across industries:

- **Move people and goods:** airlines and logistics.
- **Operate physical systems:** energy and manufacturing.
- **Allocate scarce resources:** finance and healthcare.
- **Plan and predict:** sports scheduling and machine learning.

Choose one domain

What is controlled, what is limited, and what does "best" mean?

Radiation Therapy Planning

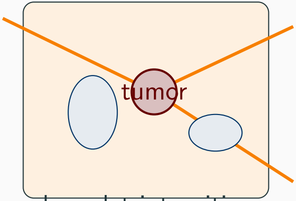
A tumor is hit by beams from many angles. Each beam is split into tiny *beamlets*; dose is proportional to intensity and doses add.

Model in words

Choose every beamlet intensity.
Minimize dose to healthy tissue.
Subject to a prescribed tumor dose, organ safety limits, and nonnegative intensities.

Flavor: linear

Proportional plus additive gives an LP with roughly 100,000 variables, solved routinely in clinics.



beamlet intensities
are decisions

Airline Crew Scheduling

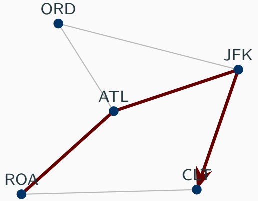
Assign crews to thousands of flights while obeying work, rest, and home-base rules.

Model in words

Choose yes or no for every candidate pairing.
Minimize total crew cost.
Subject to exactly one selected pairing covering each flight and all work/rest rules.

Flavor: discrete

One cannot fly 0.7 of a trip.
Branch-and-bound repeatedly solves LP relaxations of this integer model.



one candidate pairing: choose yes or no

Portfolio Selection

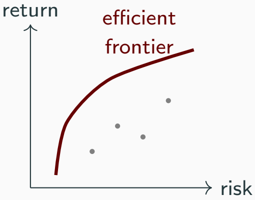
Expected return is linear in asset allocations, but variance contains products of allocations because asset returns move together.

Model in words

Choose the fraction of wealth in each asset.
Maximize expected return minus a risk penalty.
Subject to fractions summing to one and position limits.

Flavor: nonlinear

Products of variables break additivity, so the objective curves. Replacing variance by mean absolute deviation gives an LP; nonlinear algorithms also use local linear models.



The Decision-Making Maturity Curve

value ⟶

AI autonomous decision making

Optimization **THIS COURSE**
given all of that: what should we do?

Machine learning
forecasts and predictions: what will happen

Statistics patterns and why

Data what happened

sophistication ⟶

Each level consumes the levels below it. Optimization turns data, statistics, and predictions into decisions. Framing popularized by David Simchi-Levi (MIT).

You Can't Walk In and Start Optimizing

Imagine arriving at a hospital to "optimize nurse schedules." What must exist first?

1. **Data**: shift records, demand by hour, and absence rates. Every coefficient a_{ij}, b_i, and c_j must come from somewhere.
2. **Statistics and predictions**: demand forecasts, no-show probabilities, and length-of-stay estimates. Garbage in gives confidently optimal garbage out.
3. **Organizational readiness**: stable processes and people willing to act on a recommendation. A perfect schedule nobody follows optimizes nothing.

Systems engineering, process design, and data infrastructure are not detours; they often build the lower rungs that make optimization possible.

A First LP: Production Planning

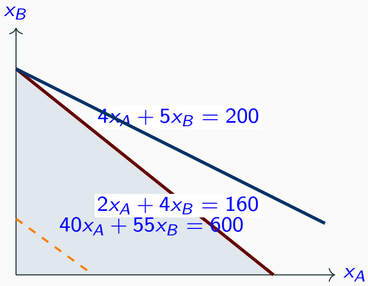
A shop makes products A and B:

	Product A	Product B	Available
Profit/unit	\$40	\$55	—
Labor (hours/unit)	4	5	200
Material (kg/unit)	2	4	160

$$\begin{aligned} \max \quad & 40x_A + 55x_B \\ \text{s.t.} \quad & 4x_A + 5x_B \leq 200, \\ & 2x_A + 4x_B \leq 160, \\ & x_A, x_B \geq 0. \end{aligned}$$

$$\max\{c^\top x : Ax \leq b,\ x \geq 0\}, \qquad A = \begin{bmatrix} 4 & 5 \\ 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 200 \\ 160 \end{bmatrix}, \quad c = \begin{bmatrix} 40 \\ 55 \end{bmatrix}.$$

Preview: The Geometry of an LP



Each constraint cuts the plane in half; what survives is the feasible region.
The objective lines $40x_A + 55x_B = t$ are parallel.
Push one in the improving direction until its last contact with the region.

Reading the graph

The last contact is the corner $(0, 40)$: make no A, make 40 units of B, and earn \$2,200.

Mathematical Programs

$$\begin{aligned}
\min / \max \quad & f(x_1, \ldots, x_n) \\
\text{s.t.} \quad & g_i(x_1, \ldots, x_n) \leq b_i, \quad i \in M_1, \\
& g_i(x_1, \ldots, x_n) \geq b_i, \quad i \in M_2, \\
& g_i(x_1, \ldots, x_n) = b_i, \quad i \in M_3, \\
& x_j \geq 0, \quad j = 1, \ldots, n.
\end{aligned}$$

Decision variables x_j quantities we control.

Constraints structural, operational, or technological limits.

Objective f the criterion to optimize.

Feasible solution a choice x satisfying every constraint.

Linear programming

The special case in which f and every g_i are linear.

LP Standard Form

$$\begin{array}{lll} \min & z = c^\top x & \text{objective} \\ \text{s.t.} & Ax = b & \text{structural constraints} \\ & x \geq 0 & \text{nonnegativity.} \end{array}$$

Definitions

Feasible region: $\{x : Ax = b,\ x \geq 0\}$. An optimal solution x^* satisfies $c^\top x^* \leq c^\top \hat{x}$ for every feasible $\hat{x}$.

$$x, c \in \mathbb{R}^n, \qquad b \in \mathbb{R}^m, \qquad A \in \mathbb{R}^{m \times n}.$$

The model data are A, b, c; the decision is x. Rows of A describe constraints and columns describe variables.

The Axioms of Linear Programming

1. **Proportionality**: each contribution is proportional to activity level, as in $c_j x_j$ and $a_{ij} x_j$.
2. **Additivity**: effects add; there are no cross-products such as $x_1 x_3$. Together, (1) and (2) give linearity.
3. **Divisibility**: fractional values such as $x_j = 2.32$ are allowed.
4. **Certainty**: all data A, b, c are known exactly.

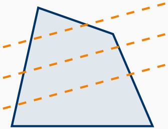
Relax an axiom, enter another field

Dropping (1)–(2) gives nonlinear programming; dropping (3) gives integer programming; modeling uncertain data gives stochastic or robust optimization. LP is the backbone on which these fields build.

Linear · Mixed-Integer · Nonlinear

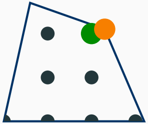
Linear

Flat sides and flat costs; an optimum can be found at a corner.



Mixed-integer

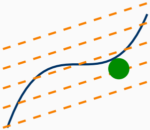
Only lattice points are allowed. Dropping integrality produces the LP relaxation and an optimistic bound.



● best integer ● LP bound

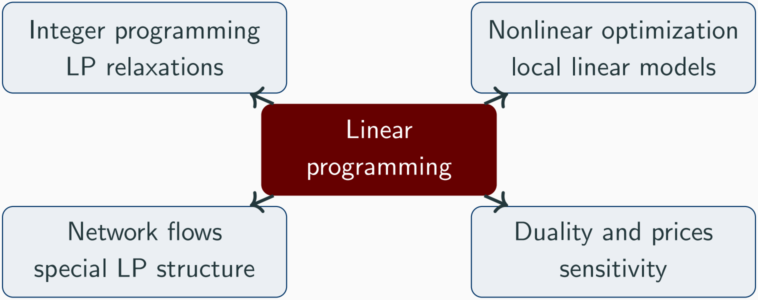
Nonlinear

Curved boundaries or costs. Tangent lines can produce LP approximations.



● curved optimum

LP is the Master Key



Branch-and-bound repeatedly solves LP relaxations; nonlinear methods use gradients and cutting planes; network models are structured LPs; duality reveals shadow prices. Master LP and you hold the key to all of them.

Why Brute Force Fails

Assign n people to n jobs, one each. There are $n!$ assignments. At 10^{12} checks per second:

n	$n!$	time to check all
10	3,628,800	about 3.6 microseconds
20	2.43×10^{18}	about 28 days
50	3.04×10^{64}	about 9.6×10^{44} years
70	1.20×10^{100}	about 3.8×10^{80} years

Conclusion

Better modeling and better algorithms beat brute force.

The Modeling Workflow

1. **Problem analysis**: understand objectives and constraints.
2. **Modeling**: translate relationships into mathematics.
3. **Model analysis**: study feasibility, convexity, and structure.
4. **Solution methods**: choose algorithms and compute solutions.
5. **Validation**: check assumptions and sensitivity, then iterate.

Real modeling is a loop

Problem analysis $\rightarrow$ model $\rightarrow$ analysis $\rightarrow$ solution $\rightarrow$ validation $\rightarrow$ revised problem. It is not a one-shot exercise.

What We Will Learn to Do

LP in general form

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & Ax \le b, \\ & A_= x = b_=, \\ & A_\ge x \ge b_\ge . \end{aligned}$$

- **Simplex family**: walk from corner to corner.
- **Duality and optimality**: certificates of best solutions and sensitivity insight.
- **Modern solvers**: Gurobi combines simplex, barrier, and presolve.

Theory and computation reinforce each other: we prove why algorithms work and use them on models in Python.

Warm-up: Is it Linear?

Classify each expression as linear as written, not allowed in an LP, or LP-representable with a reformulation.

1. $3x_1 - 2x_2 + 5x_3$: **linear**, a sum of constants times variables.
2. $x_1 x_2 + x_3$: **not an LP expression**; the product violates additivity and is not generally LP-representable.
3. $\max\{x_1, x_2\} \leq 5$: **LP-representable**; impose $x_1 \leq 5$ and $x_2 \leq 5$.
4. $\min |x_1| + 2x_2$: **LP-representable**; an auxiliary variable can model $|x_1|$.
5. $x_1 \in \mathbb{Z}$: **not an LP**; integrality violates divisibility and produces an integer program.

Preview

Lecture 2 develops the reformulations behind the two "with a trick" answers.

Materials & Tools

Primary text Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*.

References Bazaraa–Jarvis–Sherali; Dantzig–Thapa; Luenberger–Ye.

Software Gurobi with its Python interface, `gurobipy`; an academic license is free. Assignments include modeling and solving in Python.

Set up Python and `gurobipy` before the first computational assignment; a short tutorial and TA support will be available.

Homework & Late Policy

- Five graded assignments and one ungraded assignment, roughly every two weeks.
- Discuss general ideas, but write and submit your own solutions.
- Submit one organized PDF on Canvas, plus supporting code files when required.
- Content, cleanliness, and organization contribute to the grade.

Days late	Penalty
1	15%
2	30%
3	45%
4	60%

Accepted for at most four days; even one-half hour late counts as one day.

Exams & Grading

Exams

- Midterm: Thursday, October 15, during class.
- Comprehensive final: Monday, December 14, 7:45–9:45 a.m.; location announced on Canvas.
- Exceptions to the make-up policy require a university-authorized absence, documented emergency, or approved accommodation.

In-class problem solving and discussion are important practice but ungraded.

Homework	50%
Midterm	25%
Final	25%
Total	100%

AI Usage in This Course

Generative AI may support learning unless an assignment says otherwise.

Not allowed

- AI during exams.
- Complete solutions, proofs, written explanations, or code generated by AI and submitted as one's own.

Allowed support

Brainstorming, concept clarification, and code debugging, subject to each assignment's rules. Any permitted assistance contributing to submitted work must be disclosed with the tool and its use.

Students remain responsible for correctness and citations. When uncertain, ask the instructor before using the tool.

Topics Roadmap (tentative)

- Chapter 1: LP and integer-programming formulations.
- Chapter 2: polyhedra, extreme points, and linear algebra.
- Python and Gurobi modeling.
- Chapter 3: simplex, tableau, two-phase, degeneracy, revised simplex.
- Chapter 4: duality, complementary slackness, Farkas, and KKT.
- Chapter 5: sensitivity analysis.
- Variants: dual simplex and bounded variables.
- Advanced topics if time permits.

Major milestones

Homework about every two weeks; midterm October 15; final December 14. Canvas contains current assignment details.

About Your Guest Lecturer

Dr. Robert Hildebrand is an Associate Professor in Virginia Tech ISE and presents the first two meetings online.

Research

- Integer and mixed-integer nonlinear optimization.
- Game theory and multi-agent decision algorithms.
- Applied OR in redistricting, healthcare access, and disaster relief.

Background

- PhD in Applied Mathematics, UC Davis.
- Postdoctoral work at ETH Zürich, IBM Research, and the Simons Institute.
- At Virginia Tech since 2019.
- Hobbies include rock climbing and time with his children.

First-Week To-Dos

- Complete the brief introduction survey on Canvas.
- Skim Bertsimas–Tsitsiklis Chapter 1 and refresh linear algebra.
- Soon: install Python and `gurobipy` with an academic Gurobi license.
- Bookmark the course-text visualizations at https://open-optimization.github.io/open-optimization-or-book/visualizations; begin with `#modeling-intro`.

Welcome to ISE 5405!

Questions?